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Lab 6

(First half)

TASK1:

```
lab6_half.cpp
1  #include <iostream>
2  #include <fstream>
3  using namespace std;
4  struct SensorReading
5  {
6      int sensorID;
7      float temperature;
8      float humidity;
9      string timestamp;
10 };
11 int main()
12 {
13     SensorReading sensor[3] =
14     {
15         {1, 36.5, 72.3, "2024-01-15 08:00"},
16         {2, 38.1, 68.9, "2024-01-15 08:05"},
17         {3, 35.7, 75.1, "2024-01-15 08:10"}
18     };
19     ofstream file("sensor_log.txt");
20     if (file.is_open())
21     {
22         for (int i = 0; i < 3; i++)
23         {
24             file << "--- Sensor Reading " << i + 1 << " ---" << endl;
25             file << "Sensor ID : " << sensor[i].sensorID << endl;
26             file << "Temperature : " << sensor[i].temperature << " C" << endl;
27             file << "Humidity : " << sensor[i].humidity << " %" << endl;
28             file << "Timestamp : " << sensor[i].timestamp << endl;
29         }
30         file.close();
31         cout << "File \"sensor_log.txt\" created successfully." << endl;
32     }
33     else
34     {
35         cout << "Unable to open file." << endl;
36         return 0;
37     }
38 }
```

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File "sensor_log.txt" created successfully.

Reading file contents:

--- Sensor Reading 1 ---

Sensor ID : 1

Temperature : 36.5 C

Humidity : 72.3 %

Timestamp : 2024-01-15 08:00

--- Sensor Reading 2 ---

Sensor ID : 2

Temperature : 38.1 C

Humidity : 68.9 %

Timestamp : 2024-01-15 08:05

--- Sensor Reading 3 ---

Sensor ID : 3

Temperature : 35.7 C

Humidity : 75.1 %

Timestamp : 2024-01-15 08:10

Process exited after 0.2293 seconds with return value 0

Press any key to continue . . . |